

29/4.

ACTIVITY

Stack, Queue, Linked List.

2).

52
54

pop()

62
55
54

s = pop

55
54

s = 62

52
54

32
28
21

q = pop

32
28
21

q = 24

62
24
86

$$s + q = 62 + 24 = 86$$

3).

Post fix expression

10 5 + 60 6 / * 8 -

10

-

10

5

-

5

+

10 + 5

15

60

60

6

6

/

6 / 60

10

*

10 * 15

150

8

8

150 - 8

142

5
10

6
60
15

10
15

8
150

10 x 15
50
10
150

150 - 8

3. The minimum number of Enqueue operation on Q_1 required to place the elements of Q_1 in Q_2 is reverse order without using any additional storage is 0.

4. Post fix expression

8	2	3	1	/	2	3	*	+	5	1	-
8	-	8									
2	-	2									
3	-	3									
1	$2^1 3$	8									
/	$8/8$	1									
2	-	2									
3	-	3									
*	2×3	6									
+	$6+1$	5									
5	-	5									
1	-	1									

Top $\rightarrow 6$

Next = 1

5. Infix to postfix

$$a + b \times c - d \times e^1 f$$

$$= abc * + def ^ -$$

b)

$$\begin{bmatrix} 20 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 20 \\ 10 \end{bmatrix}$$

$$\begin{bmatrix} 20 \end{bmatrix}$$

poped sequence $\rightarrow 20, 20, 10, 20$